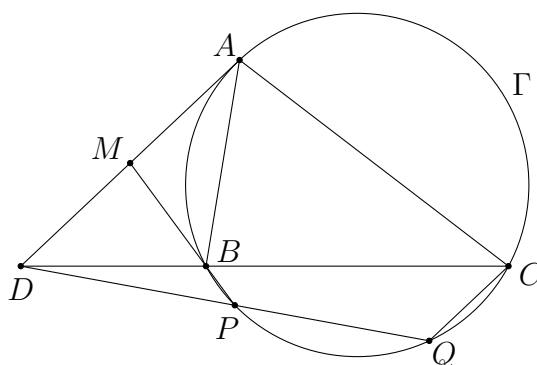


Tangents - Solutions

Problem 1. Given $\triangle ABC$, let D be the intersection of the line BC with the tangent to the circum-circle Γ of $\triangle ABC$ at A . Let M be the midpoint of AD . Let MB meet Γ again at P and let DP meet Γ again at Q . Prove that $QC \parallel DA$.

Solution.



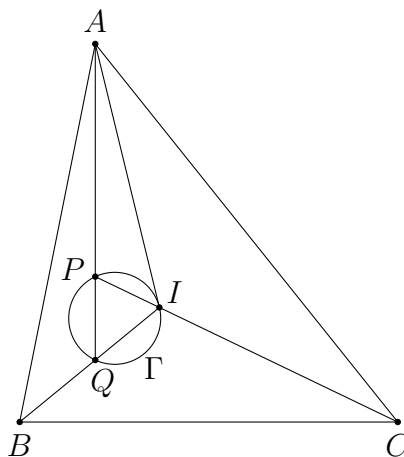
Since $MD^2 = MA^2 = MB \times MP$, MD is a tangent to (BDP) . This implies

$$\angle BDM = \angle BPD = \angle BCQ.$$

So $QC \parallel DA$.

Problem 2. In an acute scalene $\triangle ABC$, the altitude from A , the internal angle bisector of $\angle B$ and the internal angle bisector of $\angle C$ bound a triangle with circumcircle Γ . Prove that the internal angle bisector from $\angle A$ is tangent to Γ .

Solution.



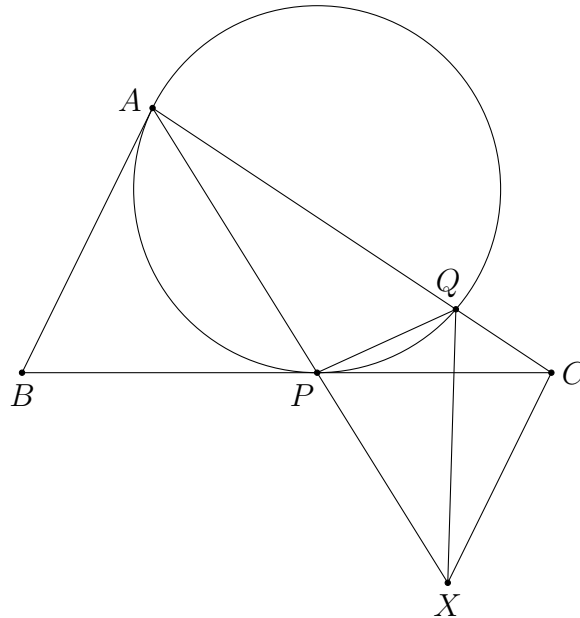
The internal angle bisectors of $\angle B$ and $\angle C$ meet at the incentre I of $\triangle ABC$. Suppose they meet the altitude from A at points Q and P respectively. Then we have

$$\angle AIP = \angle IAC + \angle ACI = \frac{A}{2} + \frac{C}{2} = 90^\circ - \frac{B}{2} = \angle BAQ + \angle QBA = \angle IQP.$$

This implies AI is tangent to Γ .

Problem 3. (Tuymaada Olympiad 2018 Junior Problem 2) A circle touches the side AB of the triangle ABC at A , touches the side BC at P and intersects the side AC at Q . The line symmetrical to PQ with respect to AC meets the line AP at X . Prove that $PC = CX$.

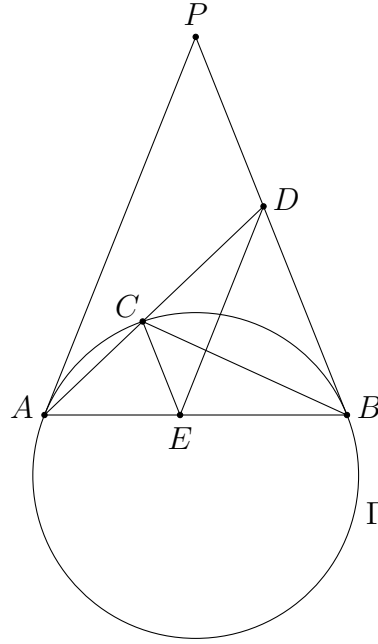
Solution.



Since PQ and QX are symmetric in the line AC , we have $\angle XQC = \angle AQP$. Also, as BP is tangent to (APQ) , we have $\angle AQP = \angle APB = \angle XPC$. This implies $\angle XQC = \angle XPC$. So C, Q, P, X are concyclic. Since $\angle CXP = \angle AQP = \angle XQC = \angle XPC$, we have $CP = CX$.

Problem 4. Let P be a point outside a circle Γ . The tangents from P to Γ touch Γ at the points A and B respectively. Let C be a point on the minor arc AB and let AC meet PB at D . Let E be the point on the segment AB such that $DE \parallel PA$. Prove that $CE \parallel PB$ if and only if $AB^2 = PB \times DB$.

Solution. Firstly, as $\angle CBE = \angle DAP = \angle CDE$, points C, E, B, D are concyclic. Then $CE \parallel DB$ if and only if $CEBD$ is an isosceles trapezoid. On the other hand, $AB^2 = PB \times DB$ if and only if BA is tangent to (PAD) , which is equivalent to $\triangle ABD \sim \triangle PBA$. As $\angle ABD = \angle PBA$ and $\triangle PAB$ is isosceles, this holds if and only if $\angle ABD = \angle ADB$, i.e. $CEBD$ is an isosceles trapezoid. This establishes the equivalence.



Problem 5. (IMO 1990 Problem 1) Chords AB and CD of a circle intersect at a point E inside the circle. Let M be an interior point of the segment EB . The tangent line at E to the circle through D , E , and M intersects the lines BC and AC at F and G , respectively. If

$$\frac{AM}{AB} = t,$$

find

$$\frac{EG}{EF}$$

in terms of t .

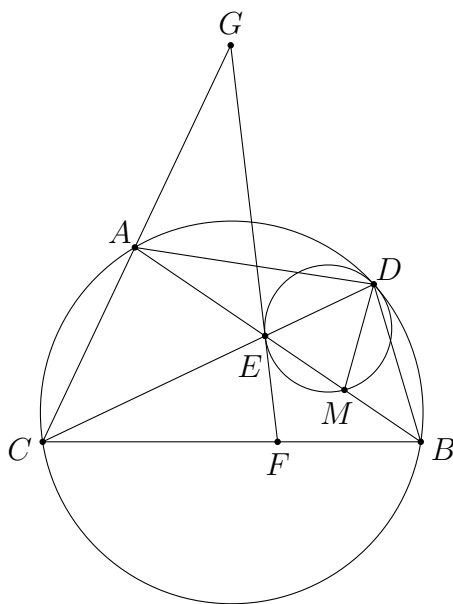
Solution. From $\angle MBD = \angle ECG$ and $\angle BMD = 180^\circ - \angle DME = 180^\circ - \angle DEG = \angle CEG$, we have $\triangle MBD \sim \triangle ECG$. This implies

$$\frac{MB}{EC} = \frac{MD}{EG}. \quad (1)$$

Similarly, from the pairs $\angle MAD = \angle ECF$ and $\angle AMD = \angle GED = \angle CEF$, we find that $\triangle MAD \sim \triangle ECF$. This implies

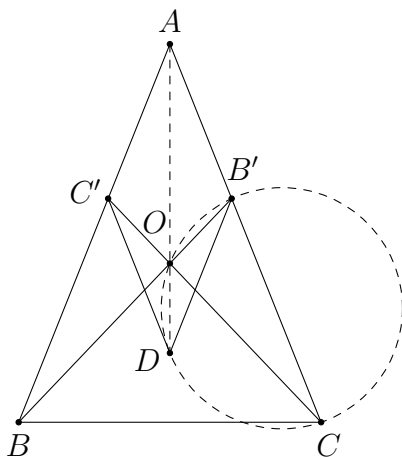
$$\frac{MA}{EC} = \frac{MD}{EF}. \quad (2)$$

Combining (1) and (2), we get $\frac{EG}{EF} = \frac{MA}{MB} = \frac{t}{1-t}$.



Problem 6. (RMO 2017 Grade 10 Problem 2) Let ABC be an acute-angled isosceles triangle with $AB = AC$ and circumcentre O . Lines BO and CO intersect AC, AB respectively at B', C' . A straight line ℓ is drawn through C' parallel to AC . Prove that the line ℓ is tangent to the circumcircle of $\triangle B'OC$.

Solution.



By symmetry, we have $AC' = AB'$. Let D be the point such that $AC'DB'$ is a rhombus. Then D lies on the axis of symmetry AO . From

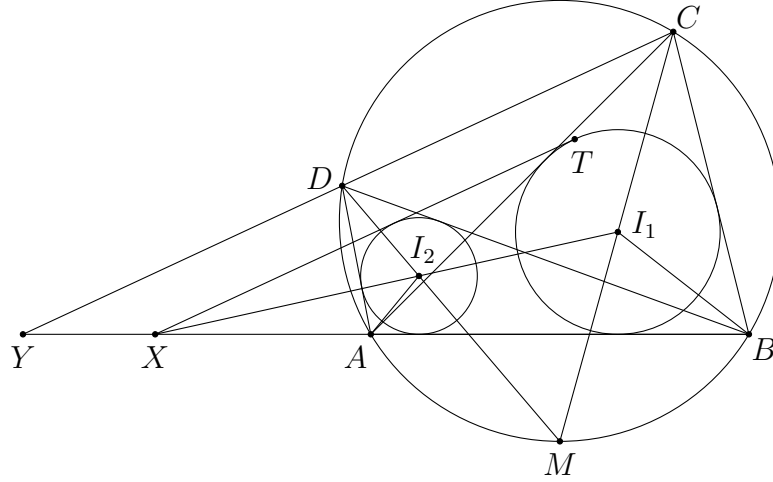
$$\angle DOC = \frac{1}{2}\angle BOC = \angle BAC = \angle DB'C,$$

we find that O, D, C, B' are concyclic. It remains to prove $C'D$ is tangent to $(ODCB')$. This is true since

$$\angle ODC' = \angle BAD = \angle OBA = \angle OB'D.$$

Problem 7. (Bosnia and Herzegovina MO 2018 Grade 12 Problem 4) Let $ABCD$ be a cyclic quadrilateral and let k_1 and k_2 be circles inscribed in triangles ABC and ABD . Prove that external common tangent of those circles (different from AB) is parallel with CD .

Solution.



Let I_1 and I_2 be the incentres of $\triangle ABC$ and $\triangle ABD$ respectively. Note that CI_1 and DI_2 meet at a point M , which is the midpoint of the arc AB . Also, as $MA = MI_2 = MI_1 = MB$, points A, B, I_1, I_2 are concyclic.

The result is obvious if $DC \parallel AB$. Assume the extensions of BA and CD meet at Y . Let I_1I_2 meet BA at X . Then X is the external homothetic centre of k_1 and k_2 . Let T be the touching point of the other external common tangent of k_1 and k_2 to k_1 . We have

$$\begin{aligned} \angle AXT &= 2\angle AXI_2 = 2(\angle BAI_2 - \angle XI_2A) = 2\angle BAI_2 - 2\angle I_1BA \\ &= \angle BAD - \angle CBA = \angle BAD - \angle YDA = \angle AYD. \end{aligned}$$

So $XT \parallel DC$.

Problem 8. (Ukrainian Geometry Olympiad 2017 Grade 10 Problem 3) On the hypotenuse AB of a right triangle ABC , we denote a point K such that $BK = BC$. Let P be a point on the perpendicular from the point K to line CK , equidistant from the points K and B . Let L be the midpoint of CK . Prove that line AP is tangent to the circumcircle of $\triangle BLP$.

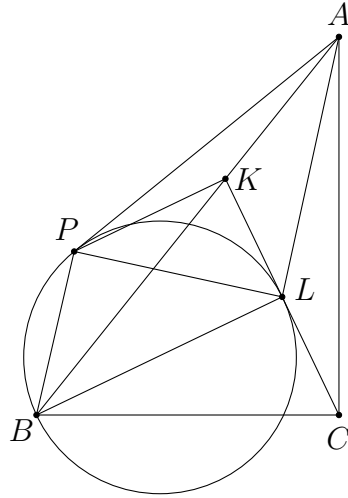
Solution. Since L is the midpoint of the base CK of the isosceles $\triangle BCK$, we have $BL \perp CK$ and hence $BL \parallel PK$. Note that

$$\angle ABP = \angle PKB = \angle LBK = \angle CBL = 90^\circ - \angle LCB = \angle ACL.$$

Also, we have

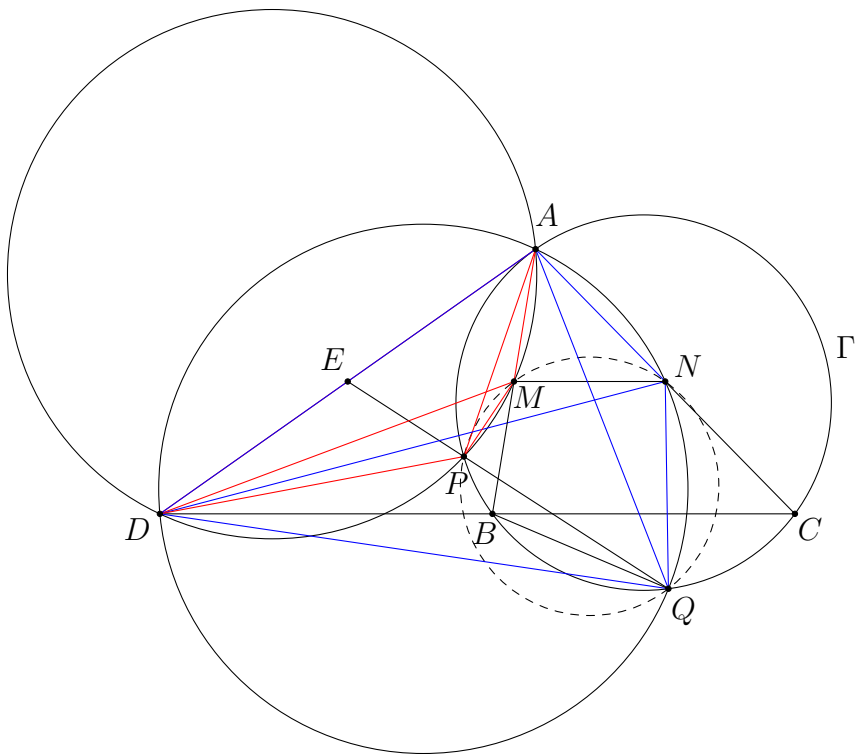
$$\frac{CL}{BP} = \frac{LK}{BP} = \sin \angle LBP = \sin \angle CBK = \frac{CA}{BA}.$$

Therefore, we have $\triangle ABP \sim \triangle ACL$. By spiral similarity, we get $\triangle APL \sim \triangle ABC$. Thus, $\angle LPA = \angle CBA = \angle LBP$. This implies AP is tangent to (PBL) .



Problem 9. Given $\triangle ABC$, let D be the intersection of the line BC with the tangent to the circumcircle Γ of $\triangle ABC$ at A . Let M and N be the midpoints of AB and AC respectively. The circumcircle of $\triangle ADM$ meets Γ again at P . The circumcircle of $\triangle ADN$ meets Γ again at Q . Prove that the line PQ bisects AD .

Solution.



Note that $\triangle ADB \sim \triangle CDA$. As M and N are the midpoints of the corresponding sides AB and CA , we have $\triangle ADM \sim \triangle CDN$. So $\angle ADM = \angle CDN$. We find that

$$\begin{aligned}
\angle QPM + \angle MNQ &= (\angle QPA - \angle MPA) + (180^\circ - \angle ANM - \angle QNC) \\
&= (\angle QBA - \angle MDA) + (180^\circ - \angle ACB - \angle QDA) \\
&= (\angle CBA + \angle QBC) - \angle CDN + 180^\circ - \angle DAB - \angle QDA \\
&= \angle CBA + \angle QAC - \angle CDN + 180^\circ - \angle DAB - \angle QDA \\
&= (\angle CDA + \angle DAB) + \angle QDN - \angle CDN + 180^\circ - \angle DAB - \angle QDA \\
&= \angle CDA + \angle QDN - \angle CDN + 180^\circ - \angle QDA \\
&= 180^\circ.
\end{aligned}$$

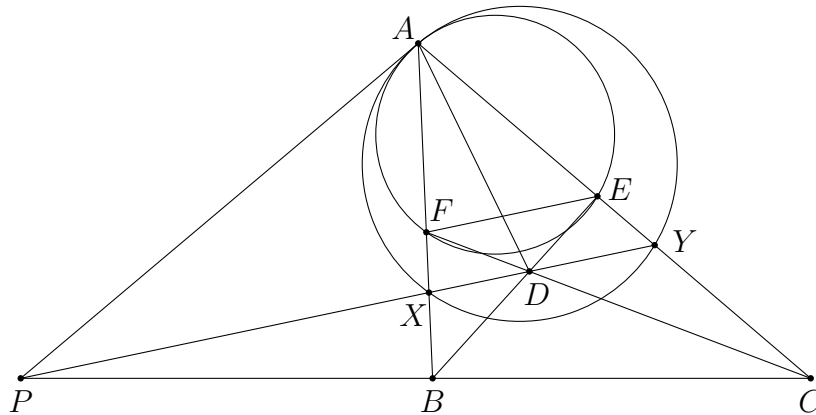
This implies N, M, P, Q are concyclic. Next, we have

$$\begin{aligned}
\angle PQD &= \angle NQD - \angle NQP \\
&= (180^\circ - \angle DAN) - (180^\circ - \angle PMN) \\
&= \angle PMN - \angle DAN \\
&= (360^\circ - \angle AMP - \angle NMA) - (\angle DAB + \angle BAC) \\
&= 360^\circ - (180^\circ - \angle PDA) - \angle CBA - \angle ACB - \angle BAC \\
&= \angle PDA.
\end{aligned}$$

This shows AD is tangent to (DQP) . Let PQ meet AD at E . Then $ED^2 = EP \times EQ = EA^2$. Thus, PQ bisects AD .

Problem 10. In $\triangle ABC$, let D be a point on the internal angle bisector of $\angle A$. The lines BD and AC meet at E . The lines CD and AB meet at F . The perpendicular bisector of AD meets the line BC at P . Prove that PA is tangent to the circumcircle of $\triangle AFE$.

Solution.



Suppose the line PD meets AB and AC at X and Y respectively. Firstly, from

$$\angle PAX = \angle PAD - \angle XAD = \angle ADP - \angle DAY = \angle AYX,$$

we find that PA is tangent to (AXY) . It suffices to show that $FE \parallel XY$, since then $\triangle AFE$ and $\triangle AXY$ are homothetic and hence PA is also tangent to (AFE) .

By Menelaus' theorem, we have

$$\begin{aligned}\frac{XP}{PY} \times \frac{YC}{CA} \times \frac{AB}{BX} &= 1, \\ \frac{AC}{CY} \times \frac{YD}{DX} \times \frac{XF}{FA} &= 1, \\ \frac{XB}{BA} \times \frac{AE}{EY} \times \frac{YD}{DX} &= 1.\end{aligned}$$

Multiplying these equations, we get

$$\frac{XP}{PY} \times \left(\frac{YD}{DX}\right)^2 \times \frac{XF}{FA} \times \frac{AE}{EY} = 1. \quad (1)$$

From $\triangle APX \sim \triangle YPA$, we have $\frac{AP}{YP} = \frac{PX}{PA} = \frac{AX}{YA}$. So

$$\frac{PX}{YP} = \frac{AP}{YP} \times \frac{PX}{PA} = \left(\frac{AX}{YA}\right)^2 = \left(\frac{DX}{YD}\right)^2$$

by the angle bisector theorem. Hence, (1) becomes $\frac{AF}{FX} = \frac{AE}{EY}$. This implies $FE \parallel XY$ and the result follows.